Ziwen Hua

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Education

09/2026 - 06/2030 **University of Maryland, College Park, MD, USA**
Undergraduate Student, Major Undeclared

09/2022 - 06/2025 **Shanghai No.2 High School, Meilong Campus, Shanghai, China**
Grade 10 - Grade 12, High School Diploma

Internship & Experience

Terps Racing, Electronics Subteam Member 09/2026 - Present

- Designed Brake System Plausibility Device (BSPD) circuit boards for TR27

Shanghai HintonAI Tech, Intern 07/2026 - 08/2026

- Designed and improved the hand-eye calibration module for a ROS2-based open source robotic painting system, RobRoss.
- Collaborated with the path-execution pipeline to ensure calibration outputs were robust in real robot runs, reducing coordinate drift and alignment errors during autonomous drawing.

Research & Projects

Research on Preparation of Rapid Cooling Equipment 09/2023 - 06/2024
Shanghai Jiaotong University

- Read papers, learned college physics and industrial cooling technology, and studied methods for physics research.
- Produced a mini semiconductor refrigerator in the lab and wrote a paper.

Application of Generative Pre-trained Transformer Models 09/2023 - 06/2024

- Researched the use of modern AI models in daily life, learned Python programming and knowledge of advanced mathematics (i.e. Bayesian Rule).
- Trained a small AI model to translate Chinese into English.

Skills & Hobbies

- **Programming:** Python, C, Java; MATLAB/Simulink for modeling and simulation
- **Robotics:** ROS2; hand-eye calibration and coordinate-frame transforms for robot arms
- **Electronics & CAD:** KiCAD schematic and PCB layout (BSPD for Formula SAE); SolidWorks part and assembly modeling
- **Machine Learning:** Self-studied "Dive into Deep Learning" (UC Berkeley textbook); core modern AI algorithms and the full model-training workflow
- **Tools:** Git/GitHub, LaTeX, Linux command line
- **Languages:** Chinese (native), English (fluent)
- **Hobbies:** Swimming, Photography

Interests

Computer Science & Engineering · Robotics · Mechanical Engineering · Racing